AHMED MAGDI Farghaly

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with **1 year of experience** specializing in **NLP**, **agentic AI workflows**, **LLMs**, and **Computer Vision**. Experienced in architecting multi-agent systems (**LangGraph**), multimodal **RAG** pipelines, and real-time edge vision solutions deployed with **Python**, **FastAPI**, **AWS**, **Streamlit**

TECHNICAL SKILLS

Agentic AI, LLMs & NLP: AI Agents, Multi-Agent Orchestration (LangGraph, LangChain), Agentic Memory, Context Engineering, RAG, Multimodal RAG, Prompt Engineering.

Computer Vision & Deep Learning: Object Detection, Multi-Object Tracking, Vision-Language Models, Diffusion Generative Models, OpenCV, CNNs, PyTorch, TensorFlow, ONNX Runtime, Edge AI (Raspberry Pi)

Backend, Databases & MLOps: Python, FastAPI, REST APIs, PostgreSQL, ChromaDB, QdrantDB, MongoDB, Vector Embeddings, Vector Search, LLM Observability & Evaluation (LangSmith, Langfuse), Cost & Latency Optimization, Pydantic, Docker, AWS, MQTT, Plotly Dash, CI/CD, Git

EXPERIENCE

Computer Vision and LLMs Trainee | NTI (National Telecommunication Institute) | Jul 2026 - Oct 2026

- Architected stateful, multi-agent reasoning workflows using LangChain and LangGraph, integrating multimodal data and Advanced Retrieval-Augmented Generation (RAG) for dynamic enterprise querying.
- Trained and optimized NLP models, including LSTMs and modern Transformers, executing complex text preprocessing and domain-specific fine-tuning on foundation base models.
- Engineered end-to-end vision systems from foundational feature extraction to deploying advanced diffusion generative models.

AI Engineer Intern | Sprint.ai | Jun 2026 - Aug 2026 | github.com/MoHatemTC/ops-chatbot-1-round-2

- Engineered a LangGraph and LLM-based escalation router, reducing manual human escalations by 40% and ensuring 100% of support tickets contained structured, actionable summaries.
- Traced LLM pipelines and graph states using LangSmith and Langfuse, enforcing strict Pydantic data validation and optimizing system latency to under 10 seconds per response.

PROJECTS

SAHM | Multi-Agent Marketing Operation System | https://github.com/Yossef-80/SAHM-AI-backend
An AI assistant that researches market, analyzes competitors, and builds brand & ad strategies for SMB brands through conversation - with persistent long-term memory of each brand.

- Architected a multi-agent system with tiered LLM intent routing and a 5-layer RAG memory architecture (session auto-summarization, long-term user profiles, and vector knowledge bases) to dynamically handle custom marketing workflows.
- Engineered a 9-tool function-calling stack for live market and competitor analysis, enforcing structured outputs with source grounding and confidence scoring to eliminate hallucinations and power interactive UI components.
- Slashed LLM inference costs and latency via watermark dirty-checking and adaptive token budgeting, implementing end-to-end observability and trace evaluations across models, tools, and databases to ensure zero data loss.

AI Lawyer Assistant | github.com/AhmedMagdi2003/ai-lawyer-assistant

- Addressed the challenge of analyzing lengthy, complex legal contracts by fine-tuning the Qwen LLM using QLoRA to achieve high-precision automated clause extraction across targeted legal topics.
- Architected a custom domain-specific RAG pipeline to ingest extensive agreements, cross-reference extracted clauses against national laws and statutory regulations, and facilitate grounded legal chat.
- Optimized model training and inference pipelines, deploying a production-ready assistant that reduced manual contract audit times and streamlined legal research workflows.

Multimodal Agentic RAG System | github.com/AhmedMagdi2003/skin-disease-multimodal-rag

- Architected a multimodal RAG application for medical skin disease analysis, unifying clinical images and diagnostic texts into a single searchable vector space using ChromaDB and SigLIP encoders.
- Engineered an autonomous agentic workflow powered by Qwen LLMs to intelligently route complex queries, analyze retrieved medical context, and generate highly accurate, grounded responses.
- Designed scalable backend data pipelines in Python to automate the ingestion, chunking, and indexing of PDFs and images, bridging complex machine learning processes with user-friendly REST APIs.

Smart Assembly Line | github.com/AhmedMagdi2003/SmartAssemblyLine

- Solved frequent physical sensor and mechanical breakdown issues by deploying an edge AI vision pipeline (YOLOv8, ONNX Runtime, BotSort on Raspberry Pi) to detect cartons, estimate orientation, and track transit times.
- Architected an MQTT-to-PostgreSQL event-logging pipeline that captures real-time conveyor telemetry and generates automated operational datasets.
- Built a Plotly Dash monitoring dashboard that eliminated slow manual tracking, enabling live system logging and generating automated analytics reports in under 5 seconds.

EDUCATION

B.Sc. Mechatronics Engineering | Assiut University | Jul 2021 - Jun 2026
Faculty of Engineering | Grade: Very Good with Honors | **GPA: 3.2 / 4.0**

CERTIFICATIONS

- **LLM Course** | Hugging Face | May 2026
- **Deep Learning Specialization** | DeepLearning.AI | Apr 2026
- **Computer Vision with PyTorch & OpenCV Bootcamp** | OpenCV | Mar 2026
- **Associate Data Scientist** | DataCamp | Aug 2025
- **McKinsey Forward Program** | McKinsey | Jul 2025
- **Data Science and AI Course** | Machinfy | Aug 2024 - Apr 2025

LANGUAGES

- Arabic: Native
- English: Intermediate (B2)
- Turkish: Conversational